

Stage-2 Hybrid Controller (AO + Temperature) – Flow Overview

Algorithm Overview:

1. Resume from previous state and reconnect all devices (TEC, AO, IDS, BME, camera).
2. Warmup phase builds models (no AO or TEC corrections).
3. In feedback mode, compute centroid (dX, dY) and OPL residual.

AO Control (Priority):

- Activates when error ≥ 0.12 px.
- Fine: 0.12–1 px.
- Assist: ≥ 1.2 px.
- Hard recovery: $\geq \sim 2$ –3 px.
- Disabled above 6 px.
- Step limits: Fine (6), Assist (10), Hard (12).
- Travel constrained to ± 22 steps (directional clamp).

Temperature Control (Secondary):

- Suppressed while AO actively correcting.
- Activated when feedback window is open and settled.
- Regimes:
 - Fine: < 2 px (~ 0.10 °C)
 - Normal: 2–3.5 px (~ 0.25 °C)
 - Recovery: > 3.5 px (~ 0.40 °C)
 - Emergency: > 5 px (~ 0.80 °C)
- Uses fused estimate (OPL + pixel + empirical dY).
- Applies PI + sign guard + quantisation.

Control Philosophy:

- AO handles fast correction.
- TEC handles slow drift.
- Both are conditionally gated for stability.

Note: This diagram is continuously updated as the controller evolves.

Start / Resume
Warmup → Model Learning (No Control)
Feedback Mode
Acquire Data → Compute dX, dY, OPL
AO Decision First
AO Apply (Fine / Assist / Hard)
TEC Decision (if allowed)
Apply Temperature Correction
Log + Save State → Loop